

Project Title: SecureSwitch

**Course: Bachelor of Computer Applications
(BCA)**

Institution: Seshadripuram Degree College

Project Group Members:

- **Shaun Kumar**
- **Liran Kaverappa K J**
- **Thejus Kumar N**
- **Mayur K**
- **Manoj M**
- **Manjunath K**

Hardware Platform: Arduino Uno & HC-SR04 Ultrasonic Sensor

Visualization Platform: Processing Development Environment (Java)

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Abstract

This project presents an integrated IoT system that combines hardware sensing with advanced graphical visualization. By utilizing an ultrasonic sensor, the team developed a "touchless" interface featuring a Distance Radar, an F1-style Reaction Timer, and a Flappy Plane Game.

To overcome the limitations of small, fragile hardware screens, the project employs Serial Data Visualization to render a high-definition Neon Dashboard. The system is secured by a distance-based gesture password, ensuring that only authorized users can access the applications.

Introduction

The core objective of this project is to explore Human-Computer Interaction (HCI) using non-contact sensors. In a modern world where touchless technology is increasingly important, this system demonstrates how simple sound waves (ultrasonic) can be used to navigate complex menus and control real-time games. By bridging an Arduino Uno with a laptop workstation, the project showcases the power of Serial Communication in creating professional-grade IoT displays.

System Overview

The system operates in four distinct software modes:

- 1. Secure Lock: A default state requiring specific hand-distance sequences to unlock.**
- 2. Distance Radar: A precision utility that visualizes proximity with a dynamic bar and numerical readout.**
- 3. Reaction Timer: A competitive reflex tester that uses F1-inspired starting lights and color-coded results.**

- 4. Flappy Plane: A physics-based game where hand height controls flight, featuring increasing difficulty (speed scaling) and high-score tracking.**
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Components and Materials Used

- **Arduino Uno:** The primary microcontroller responsible for sensor data acquisition.
 - **HC-SR04 Ultrasonic Sensor:** Used to measure distance by emitting high-frequency sound pulses.
 - **Laptop (PC):** Serves as the primary 1280x720 HD display and processing hub for graphics.
 - **USB Data Cable:** Used for the Serial data link between the hardware and software.
 - **Processing Software:** A Java-based IDE used to create the Neon Dashboard interface.
 - **Jumper Wires & Breadboard:** For hardware circuit assembly.
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System Architecture

The project follows a Client-Server Architecture via Serial Bridge:

- **Data Collection (Arduino):** The sensor measures the distance; the Arduino packages this as a string (e.g., "D:25").

- **Communication:** Data is sent over a USB Serial link at a baud rate of 9600.
 - **Graphical Rendering (Processing):** The laptop "listens" to the incoming data, parses the numbers, and updates the visual elements of the dashboard at 60 frames per second.
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Working of the Project

The project relies on the Time-of-Flight (ToF) principle. The ultrasonic sensor sends a pulse that bounces off a hand; the Arduino calculates the distance based on the echo time.

- **Sticky Logic:** To ensure security, the system uses a "Success Counter." A user must hold their hand steady at a target distance for 25 consecutive cycles to validate a password step.
 - **Dynamic UI:** On the home screen, a blue bar moves in real-time as the user's hand moves, providing instant visual feedback.
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Implementation

The project was implemented using two coding environments:

- **Arduino (C++):** Optimized for sensor polling and low-latency data transmission.

- **Processing (Java):** Used for HD rendering, neon color palettes, and game physics.
 - **Performance Logic:** The Reaction Timer displays results in Neon Green if the user reacts in $\leq 300\text{ms}$ and Neon Red if they are slower, providing immediate performance analysis.
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Results and Output

- **Enhanced Visualization:** Moving the output to a laptop screen solved the issue of hardware display fragility while providing 10x the screen resolution.
 - **Reliability:** The smoothing filter (lerp) successfully eliminated sensor "flicker," making the Flappy Plane and Distance Radar very stable.
 - **Gamification:** The Flappy Plane successfully increases speed every 10 points, providing a challenging and polished gaming experience.
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Conclusion

The SecureSwitch successfully met all its design objectives. It proves that low-cost hardware like the Arduino can be combined with modern laptops to create high-end, secure applications. This project serves as a strong foundation for future work in touchless interfaces and remote sensor

monitoring, which are critical skills in the field of Computer Applications.

References

- 1. NewPing Library: Optimized ultrasonic sensing logic.**
- 2. Processing.org Documentation: Serial communication and 2D shape rendering.**
- 3. Arduino Language Reference: Serial communication protocols and mapping functions.**